

Cover Page

COMP5047: Applied Software Engineering

Resit Coursework Submission

Subsystem: CloudTables-Manager

Student Number: 19268438

(No name included as per specification)

1. Project Management and SE Tools

GitHub Repository URL: <https://github.com/Faheem9798/cloudtable-resit.git>

UML Modelling Tool Used: Lucidchart

List of Files on GitHub:

- use_case_diagram.png
- activity_diagram.png
- component_diagram.png
- class_diagram.png
- sequence_diagram.png
- lucidchart source files
- submission.pdf (final coursework document)

Introduction

The CloudTables-Manger Subsystem is a web application designed for managers to run their operations efficiently. It simplifies tasks such as managing reservations, organizing staff, monitoring inventory, analyzing performance etc, As an essential part of the cloud system. It works seamlessly with the customer, service and operations subsystems to create an efficient restaurant management solution. Built using a microservices architecture. This subsystem prioritises scalability, reliability and performance while adhering to strict security standards.

Subsystem Features

Reservation Management

- **Security:** Managers only, 2FA login.
- **Performance:** Process updates in under 2 seconds
- **Reliability:** 99.9% uptime target.
- **Scalability:** Handle 50+ users concurrently.

Staff Scheduling

- **Security:** Restricted to manager accounts.
- **Performance:** Display/update schedules instantly.
- **Reliability:** Notify on conflicts.
- **Scalability:** Support 10+ branches.

Inventory Monitoring

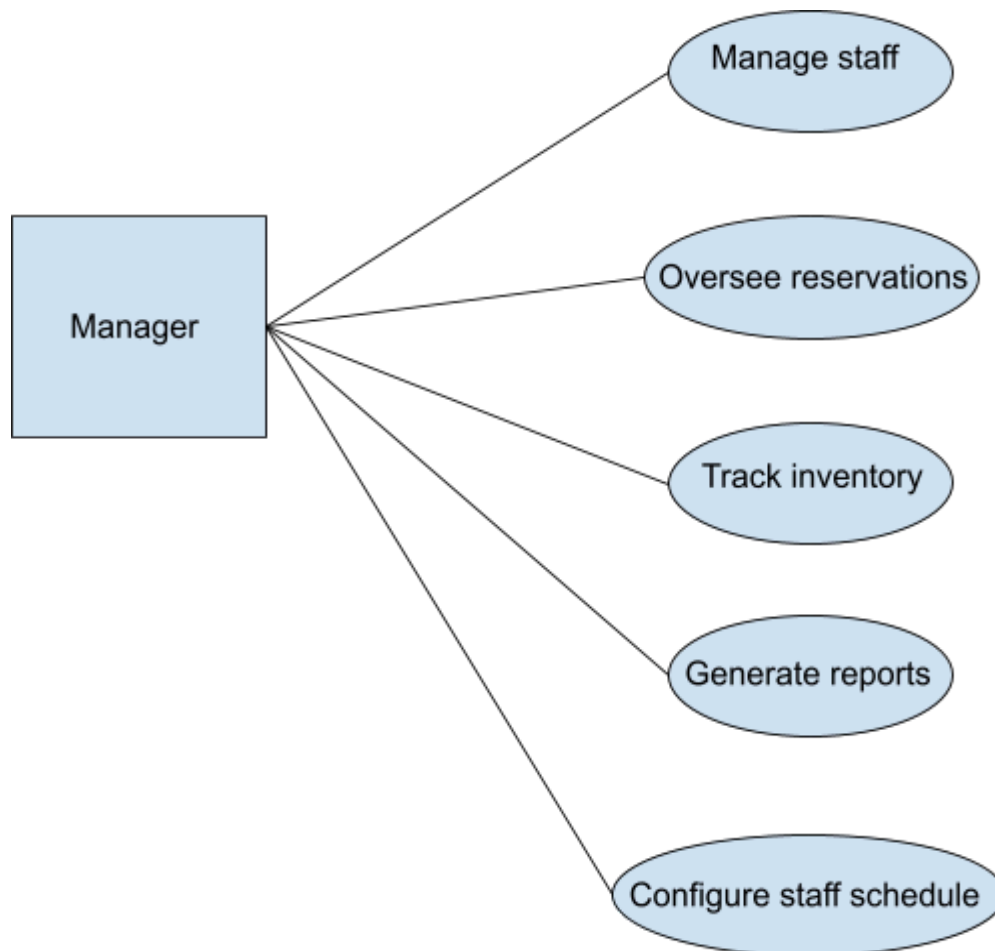
- **Security:** Inventory manager role only.
- **Performance:** Real-time alerts within 1s.
- **Reliability:** Auto-backup and recovery.
- **Scalability:** Up to 500+ items.

Performance Analytics

- **Security:** Read-only access via encrypted reports.
- **Performance:** Render reports in under 5s.
- **Reliability:** Archive historical reports.
- **Scalability:** Handle annual branch data.

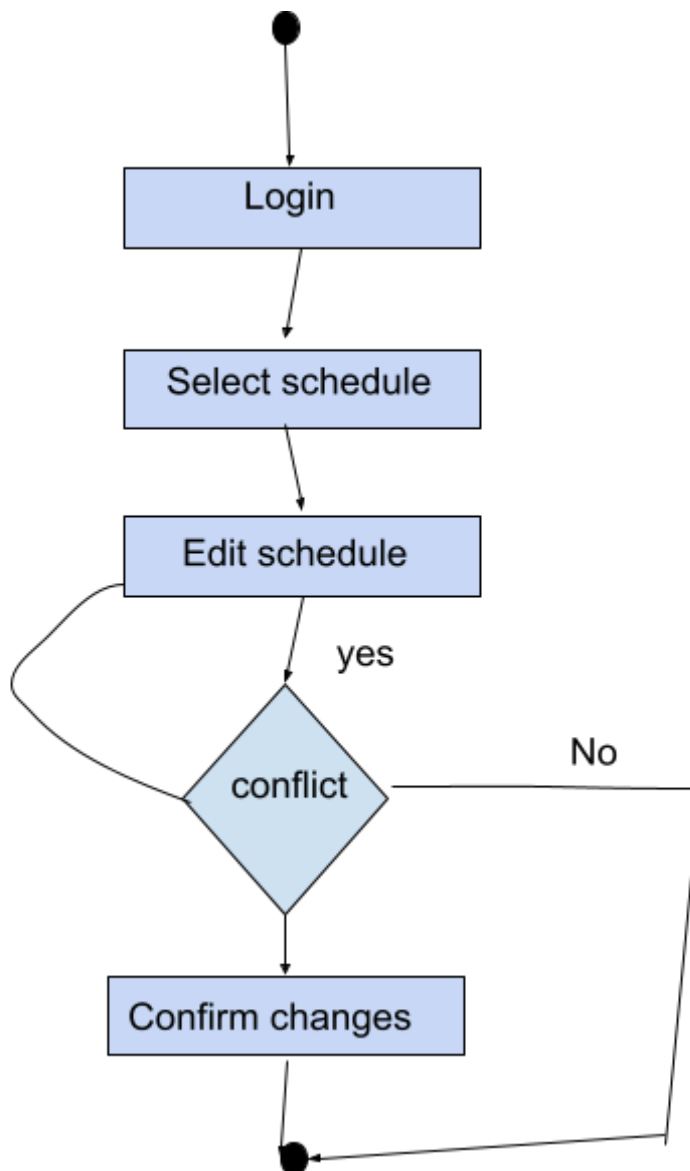
Functional Model

Use Case Diagram



This use case diagram model represents the core functionalities available to managers: managing reservations, configuring staff schedules, inventory monitoring, performance analysis, and managing accounts.

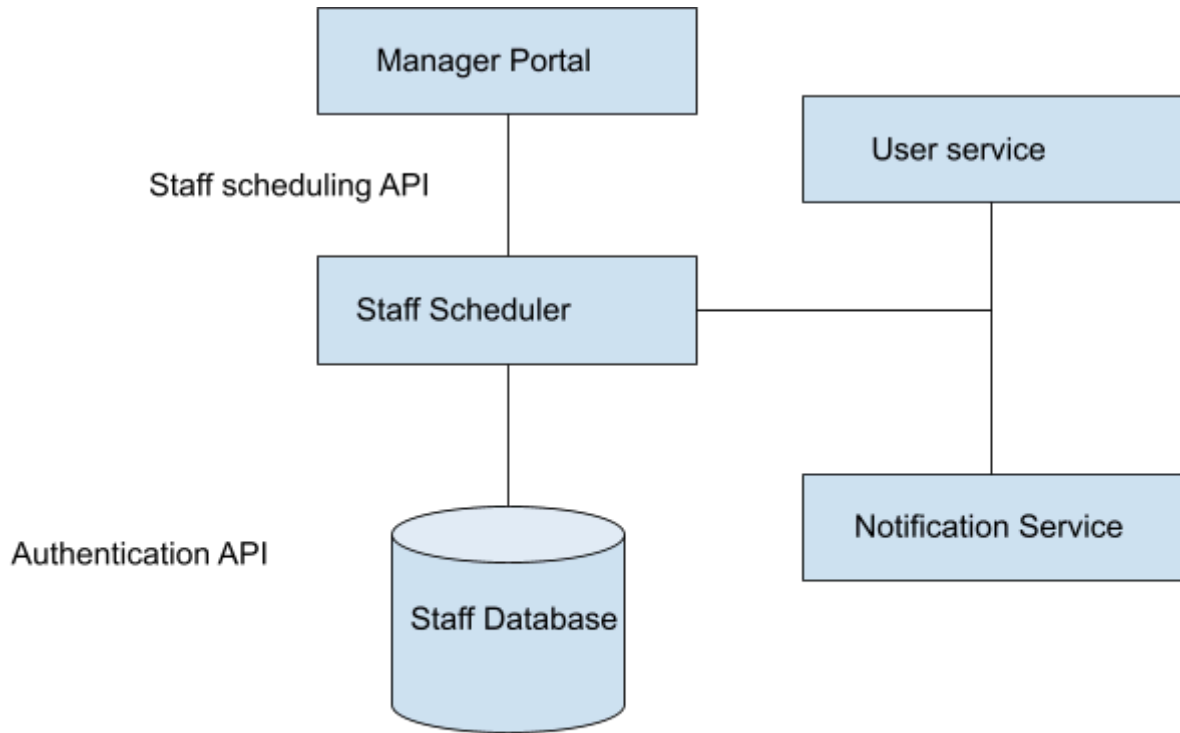
Activity Diagram: Configure Staff Schedules



This activity diagram outlines the manager process to access staff shifts and to edit them it also shows how the systems acts in a situation of conflict and how it redirects once the conflict is resolved.

Architectural Design

UML Component Diagram



The component diagram presents a modular architecture based on microservices, showing how the ManagerPortal interacts with services like StaffScheduler, UserService, NotificationService, and the central Staff Database.

4.2 Component Specifications

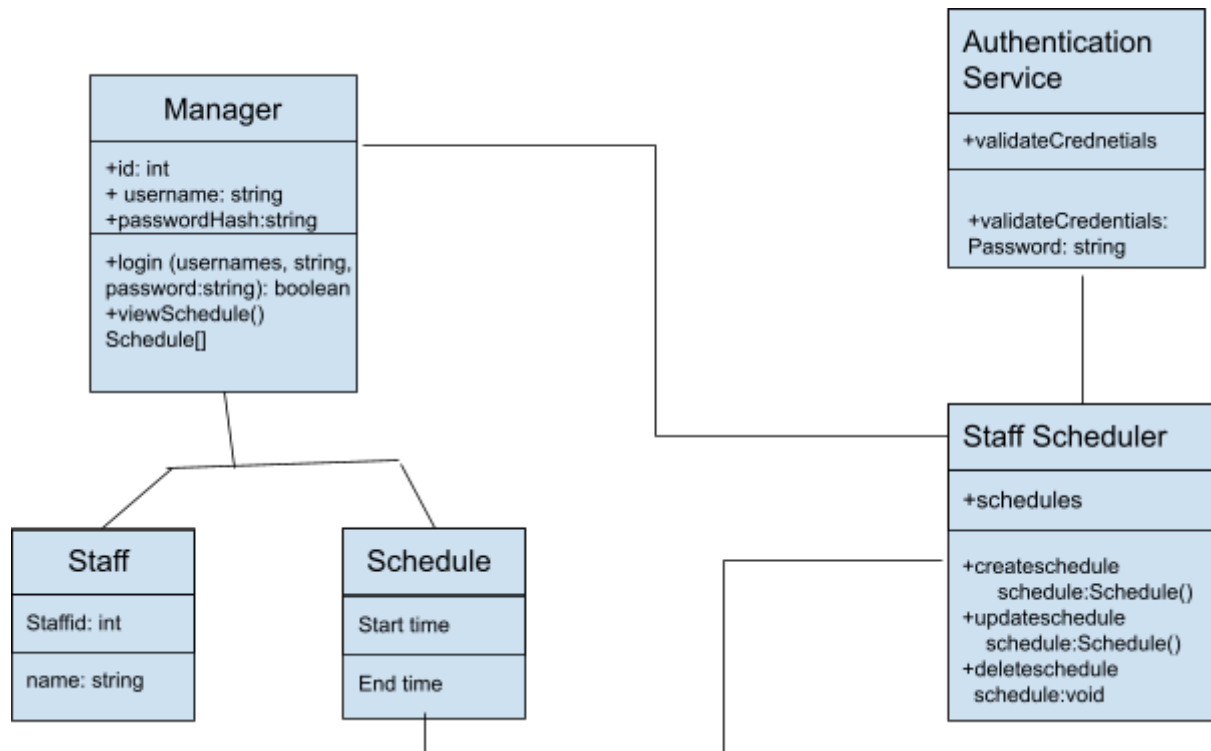
- **ManagerPortal:** Frontend interface for manager users.
- **StaffScheduler:** Main logic component for managing and validating schedules.
- **UserService:** Handles authentication and role validation.
- **NotificationService:** Sends alerts and notifications.
- **Staff Database:** Secure data store for all scheduling and staff information.

Interfaces:

- `/login(), /createSchedule(), /notify(), /validateSchedule()`
-

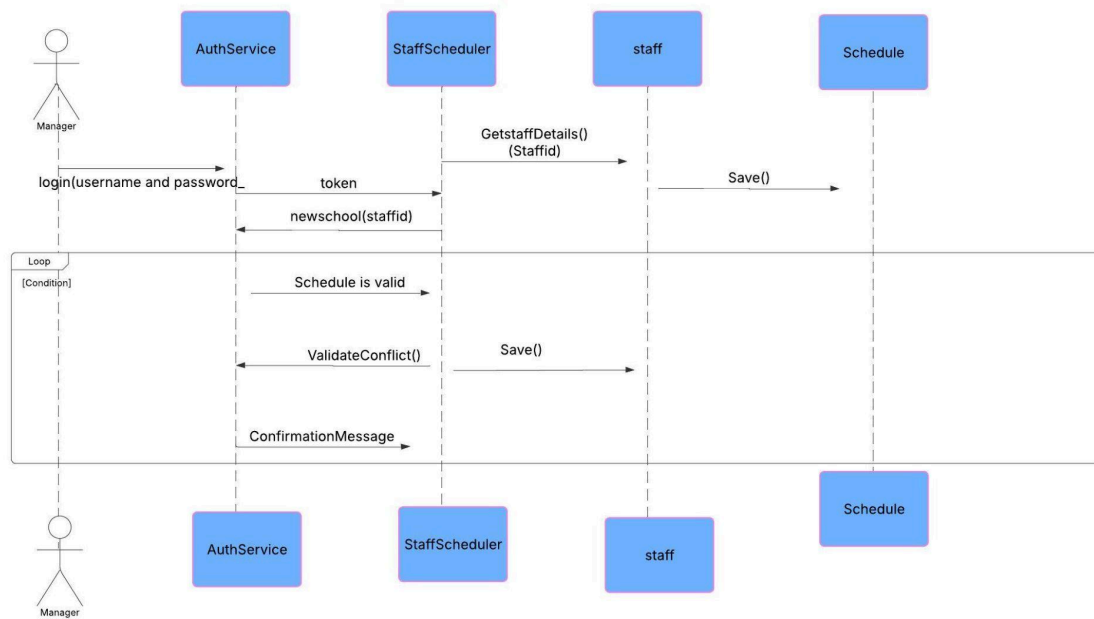
Detailed Design

Class Diagram



This class diagram models the internal structure of the **StaffScheduler** subsystem, including its relationship with **Manager**, **AuthService**, **Schedule**, and **Staff**. It includes methods for login, schedule creation, update, and deletion.

UML Sequence Diagram



This sequence diagram visualizes the process of a manager logging in and creating a new schedule. It shows authentication via **AuthService**, interaction with **StaffScheduler**, staff validation, conflict checking, and confirmation handling. An **alt** fragment is used to represent the conditional flow for valid vs. invalid schedules.

